

Low Cardinalities of Quantum Latin Squares of Order Five: Computational Certificates for the Gaps at 8, 9, 10, and 11

Research notes generated from the QLS spectrum project

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Abstract

We record the current small-order evidence for the cardinality spectrum of quantum Latin squares. The main new computational conclusion is that no quantum Latin square of order five has cardinality 8, 9, 10, or 11. Together with the general exclusion of cardinality $n + 1$, and with explicit constructions of cardinalities 5, 7, 12, this gives

$$\{5, 7, 12\} \subseteq \text{Spec}(5), \quad 6, 8, 9, 10, 11 \notin \text{Spec}(5).$$

The exclusions of 8, 9, 10, and 11 are finite, reproducible, computer-assisted classifications. The final obstruction checks are ordinary two- and three-dimensional linear algebra. We also record an exact cardinality-12 construction and the next target $c = 13$.

1 Definitions and context

A quantum Latin square of order n , abbreviated $\text{QLS}(n)$, is an $n \times n$ array $Q = (q_{ij})$ whose entries are unit vectors in $\mathbb{C}^n$, such that each row and each column is an orthonormal basis of $\mathbb{C}^n$. Two entries are identified if they differ by a nonzero scalar; in the unit-vector normalization this means they differ by a global phase. The cardinality of Q is the number of distinct projective classes appearing in the array.

Let $\text{Spec}(n)$ denote the set of possible cardinalities of $\text{QLS}(n)$. The trivial bounds are

$$n \leq c \leq n^2.$$

Classical Latin squares give $c = n$. General constructions give maximal cardinality n^2 for all $n \geq 4$. Zhang–Wang–Ji proved that for all multiples $n = 4m$, $m \geq 2$,

$$\text{Spec}(n) = [n, n^2] \setminus \{n + 1\}.$$

They also proved the following general obstruction.

Lemma 1 (Zhang–Wang–Ji). *For every $n \geq 2$, no $\text{QLS}(n)$ has cardinality $n + 1$.*

Thus $6 \notin \text{Spec}(5)$. The remaining low-cardinality question for order five begins at $c = 7, 8, 9, \dots$

2 Main order-five conclusions

Theorem 1 (Certified low spectrum of order five). *The current certified low-cardinality data for order five is*

$$\{5, 7, 12\} \subseteq \text{Spec}(5), \quad 6, 8, 9, 10, 11 \notin \text{Spec}(5).$$

Proof. The value 5 is classical. The value 7 is obtained by rotating a single intercalate in a classical Latin square of order five. The value 12 is given explicitly in Section 8. The exclusion of 6 is the $n + 1$ lemma. The exclusions of 8 and 9 are the finite certificates described in Sections 4 and 5; the exclusion of 10 is described in Section 6; the exclusion of 11 is described in Section 7. $\square$

Corollary 1. *The naive extension*

$$\text{Spec}(n) = [n, n^2] \setminus \{n + 1\} \quad \text{for all } n \geq 5$$

is false already at $n = 5$.

3 Normalized pattern method

Let Q be a hypothetical QLS(5) of small cardinality. Applying a global unitary, we may assume that the first row is

$$e_0, e_1, e_2, e_3, e_4.$$

The old projective classes are labelled 0, 1, 2, 3, 4, corresponding to the coordinate vectors. If the cardinality is $5 + t$, the remaining classes are labelled $5, \dots, 4 + t$. A cell below the first row is called active if it contains a new label.

No label can repeat in a row or column. In addition, old label j cannot occur below the first-row entry in column j . For a new label k , define

$$Z_k = \{j \in \{0, 1, 2, 3, 4\} : x_k \perp e_j \text{ is forced}\}, \quad S_k = \{0, 1, 2, 3, 4\} \setminus Z_k.$$

The vector x_k must lie in $\text{span}\{e_j : j \in S_k\}$. Since x_k is not a coordinate vector, one must have

$$|S_k| \geq 2.$$

Whenever two new labels occur in a common row or column, their vectors must be orthogonal. This gives an orthogonality graph on the new labels.

4 The gap at cardinality 8

Theorem 2. *There is no QLS(5) of cardinality 8.*

Proof. After the first-row normalization there are three new labels, 5, 6, 7. The deterministic classifier enumerates all admissible normalized label patterns and records the support sets S_5, S_6, S_7 and the forced orthogonality graph. It gives the following exhaustive classification.

Obstruction type	Patterns	Explanation
<code>same_2d_path</code>	8640	three vectors in one 2-space with a path
<code>same_2d_triangle</code>	23040	three mutually orthogonal vectors in one 2-space
<code>two_2d_one_3d_triangle</code>	2880	two vectors form a 2-space basis; the third degenerates

The full certificate statistics are

$$\begin{aligned} \text{unique patterns} &= 34560, \\ \text{unobstructed patterns} &= 0, \\ \text{SHA-256 digest} &= 8837bc4dd3c92ab424d30c4f1be7f20f \\ &\quad 235de8e87cee6563d6c202a609ed1a75. \end{aligned}$$

The three obstruction types are impossible by elementary linear algebra. First, a two-dimensional Hilbert space cannot contain three mutually orthogonal nonzero projective vectors. Second, if $x_5 \perp x_6$ and $x_6 \perp x_7$ inside the same two-dimensional subspace, then x_5 and x_7 lie in the same one-dimensional orthogonal complement of x_6 , so they are projectively equal. Third, if two vectors form an orthogonal basis of a two-dimensional support S , then any vector orthogonal to both has zero projection onto S ; if its support is $S \cup \{t\}$, it is forced onto the coordinate line $\mathbb{C}e_t$, so it is not a new projective class. $\square$

5 The gap at cardinality 9

Theorem 3. *There is no QLS(5) of cardinality 9.*

Proof. There are four new labels, 5, 6, 7, 8. The active-cell graph has four row vertices, five column vertices, and an edge for each active cell. The script `qls5_active_graph_general.py --t 4` enumerates all 2^{20} active graphs up to row and column permutations and applies the basic line-degree, old-label fillability, new-label colorability, and local support conditions. The counts are:

$$\begin{aligned} \text{raw graphs} &= 1048576, \\ \text{line-degree feasible types} &= 151, \\ \text{old-fill feasible types} &= 151, \\ \text{new-coloring feasible types} &= 150, \\ \text{old-and-new feasible types} &= 150, \\ \text{local-support feasible types} &= 14. \end{aligned}$$

Thus only 14 active graph types survive.

The deterministic certificate then expands these 14 graph types into all normalized old/new label fillings. It finds

$$\begin{aligned} \text{label patterns} &= 1587, \\ \text{support-edge types} &= 97, \\ \text{unclassified support-edge types} &= 0. \end{aligned}$$

All 97 support-edge types fall into the following three deterministic obstruction classes.

Obstruction type	Support-edge types	Label patterns
<code>clique_union_dim</code>	47	1451
<code>same_2d_basis_common_neighbor</code>	33	44
<code>same_2d_two_step</code>	17	92

The first obstruction says that a clique of q mutually orthogonal nonzero vectors cannot be supported inside a coordinate subspace of dimension less than q . The second says that if two adjacent vectors have the same two-dimensional support, they form a basis of that support; a common neighbor must therefore be supported outside that two-space, and if fewer than two outside coordinates remain, it is zero or a coordinate vector. The third is the two-step path obstruction already used for $c = 8$. Since every remaining support-edge type is obstructed, no cardinality-9 square exists. $\square$

6 The gap at cardinality 10

Theorem 4. *There is no QLS(5) of cardinality 10.*

Proof. There are five new labels, 5, 6, 7, 8, 9. The active graph sieve starts with all 2^{20} active-cell graphs and applies the same line-degree, old-label fillability, new-label colorability, and local

support constraints. The counts are

$$\begin{aligned}
&\text{raw graphs} &= 1048576, \\
&\text{line-degree feasible types} &= 222, \\
&\text{old-fill feasible types} &= 222, \\
&\text{new-coloring feasible types} &= 220, \\
&\text{old-and-new feasible types} &= 220, \\
&\text{local-support feasible types} &= 40.
\end{aligned}$$

Thus only 40 active graph types survive the combinatorial and local support filters.

The deterministic certificate expands those 40 graph types into all normalized old/new label fillings and collapses them to support-edge types. It finds

$$\begin{aligned}
&\text{label patterns} &= 9010, \\
&\text{support-edge types} &= 1724, \\
&\text{unclassified support-edge types} &= 0.
\end{aligned}$$

The support-edge types are distributed as follows.

Obstruction type	Support-edge types	Label patterns
<code>clique_union_dim</code>	1064	8052
<code>two_2d_overlap_one_edge</code>	601	817
<code>same_2d_two_step</code>	46	116
<code>same_2d_basis_common_neighbor</code>	13	25

The first, third, and fourth obstruction classes are those used for cardinality 9. The new class is `two_2d_overlap_one_edge`: if two adjacent vectors have two-dimensional supports S_u, S_v with $|S_u \cap S_v| = 1$, then their inner product has only one possible common coordinate contribution. Hence orthogonality forces at least one vector to vanish on the common coordinate, so that vector is supported on a single coordinate line and is not a new projective class. Since all 1724 support-edge types are obstructed, no cardinality-10 square exists. $\square$

7 The gap at cardinality 11

Theorem 5. *There is no QLS(5) of cardinality 11.*

Proof. There are six new labels after the first-row normalization. The fast active-graph layer gives 220 old-and-new feasible active graph types. Instead of using the older generic local-support prefilter, the incremental certificate enumerates label fillings directly over these 220 graph types, enforcing $|Z_k| \leq 3$ during assignment and introducing new labels in canonical first-occurrence order.

The complete enumeration gives

$$\begin{aligned}
&\text{label patterns} &= 201297, \\
&\text{support-edge types} &= 46808.
\end{aligned}$$

After reclassification with the final obstruction library, all support-edge types are obstructed:

$$\text{unclassified support-edge types} = 0.$$

The final counts are

Obstruction type	Support-edge types	Label patterns
<code>clique_union_dim</code>	36853	184669
<code>two_2d_overlap_one_edge</code>	9804	16346
<code>same_2d_two_step</code>	131	262
<code>same_2d_basis_common_neighbor</code>	18	18
<code>same_3d_two_independent_common_neighbors</code>	2	2

The last obstruction is new at $c = 11$. Suppose two distinct new vectors u, w have the same three-dimensional support S , and suppose they have two common neighbors a, b whose two-dimensional supports are contained in S , meet in one coordinate, and together cover S . Then a and b are necessarily linearly independent unless one degenerates to a coordinate vector. Therefore the orthogonal complement of $\text{span}(a, b)$ in $\text{span}(S)$ is one-dimensional. Both u and w lie in this line, so they are projectively equal, contradicting that they are distinct new classes. This eliminates the final two support-edge types; hence no cardinality-11 square exists. $\square$

8 An exact cardinality-12 construction

Let e_i be the standard coordinate vectors of $\mathbb{C}^5$. Define the following unnormalized vectors:

$$\begin{aligned}
a &= (0, -1, 0, 2, 1), & b &= (0, 1, 0, 1, -1), & c &= (0, 1, 0, 0, 1), \\
d &= (0, -1, 0, 2, -5), & h &= (0, 2, 0, 1, 0), & f &= (0, 1, 0, -2, 0), \\
g &= (0, 1, 0, 0, -1).
\end{aligned}$$

After normalizing these seven vectors, the array

$$\begin{array}{ccccc}
e_0 & e_1 & e_2 & e_3 & e_4 \\
a & e_0 & b & c & e_2 \\
d & e_2 & a & e_0 & h \\
h & e_4 & e_0 & e_2 & f \\
e_2 & e_3 & c & g & e_0
\end{array}$$

is a QLS(5) of cardinality 12. Verification is exact: all row and column orthogonality checks reduce to integer dot products before normalization. The corresponding verifier is the script `exact_qls5_c12_construction.py`.

Remark 1. A tangent-space test around this construction was also performed. With the first row fixed, the base construction has projective tangent nullity 4. For every tested single-cell split of a repeated class below the first row, the projective tangent nullity remains 4. Thus this construction appears first-order rigid against the simplest local splitting route toward cardinality 13. This is not a nonexistence theorem for $c = 13$.

9 Next target

The next unresolved low value is $c = 13$. The beginning of the order-five spectrum now has the certified jump

$$5, 7, 12, \dots$$

The known cardinality-12 construction is first-order rigid against simple single-cell splitting, so the next target likely requires either a more global modification of the $c = 12$ pattern or a different construction altogether.

The first active-graph layer for $c = 13$ has 216 old-and-new feasible graph types. Incremental enumeration has currently completed graph types 1 through 194, with several million-pattern graph types already processed. Sampled large graph types, including one with 1,092,990

support-edge types, are fully covered by the current obstruction library. However, the complete $c = 13$ classification is not yet finished: the bottleneck is now data engineering, since some individual graph certificates are multi-gigabyte-scale if every support-edge type is written to JSON. The next version should classify support-edge types online and save only obstruction counts, samples of unclassified types, and reproducibility hashes.

10 Reproducibility

The main files produced in this stage are listed below.

File	Role
<code>c8_classification_certificate.py</code>	deterministic cardinality-8 classifier
<code>c8_classification_certificate.json</code>	compact cardinality-8 certificate
<code>qls5_active_graph_general.py</code>	active-graph sieve for $c = 8, 9, 10$
<code>qls5_c9_deterministic_certificate.py</code>	proof-oriented cardinality-9 certificate
<code>qls5_c9_deterministic_certificate.json</code>	cardinality-9 certificate output
<code>qls5_c9_support_obstruction.py</code>	support-edge obstruction classifier
<code>qls_support_obstruction_general.py</code>	general support-edge obstruction classifier
<code>qls5_c10_deterministic_probe.py</code>	proof-oriented cardinality-10 certificate
<code>qls5_c10_deterministic_probe_complete.json</code>	cardinality-10 certificate output
<code>qls5_incremental_probe.py</code>	canonical incremental probe used for $c = 11$
<code>qls5_c11_incremental_old_and_new_ok_complete_reclassified.json</code>	cardinality-11 certificate output
<code>qls5_c13_progress_note.md</code>	current cardinality-13 progress report
<code>reclassify_support_certificate.py</code>	certificate reclassification utility
<code>exact_qls5_c12_construction.py</code>	exact cardinality-12 verifier
<code>tangent_split_c12.py</code>	first-order split test around the $c = 12$ construction